

NURS 1020 Anatomy and Physiology 2

Physiology Assignment

Due date: March 8, 2026, at 2359

Submit via the Physiology Assignment link on Blackboard.

Instructions:

- APA formatting must be used for the title page, in-text citations, and the reference list at the end.
- **The maximum number of type-written pages is ten (10) pages including the title page and the reference list.**
- The assignment should be devoted to answering the questions related to the individual experiments. Place the question number (e.g., Q1.) and then type your discussion to that question. Please do not duplicate the question itself.
- Please refer to the associated evaluation rubric.
- Submit the assignment through the same link where you downloaded the assignment from.
- Due Date for Submission of Physiology Assignment: March 8, 2026, at 2359
- **Please note that this assignment is an academic paper. Your content must be cited to sources (e.g., textbook, peer-reviewed article, evidence-based health website, etc.).**
- **Please also note that deductions for APA formatting errors will follow the guidelines in the HBSN program handbook. See attached rubric for further information.**

Grading Scheme

Virtual Physiology Experiment 1:	15 marks
Virtual Physiology Experiment 2:	15 marks
Virtual Physiology Experiment 3:	15 marks

Total: 45 marks (which will be converted to 15% of the final course grade)

Please email me if you have any questions.

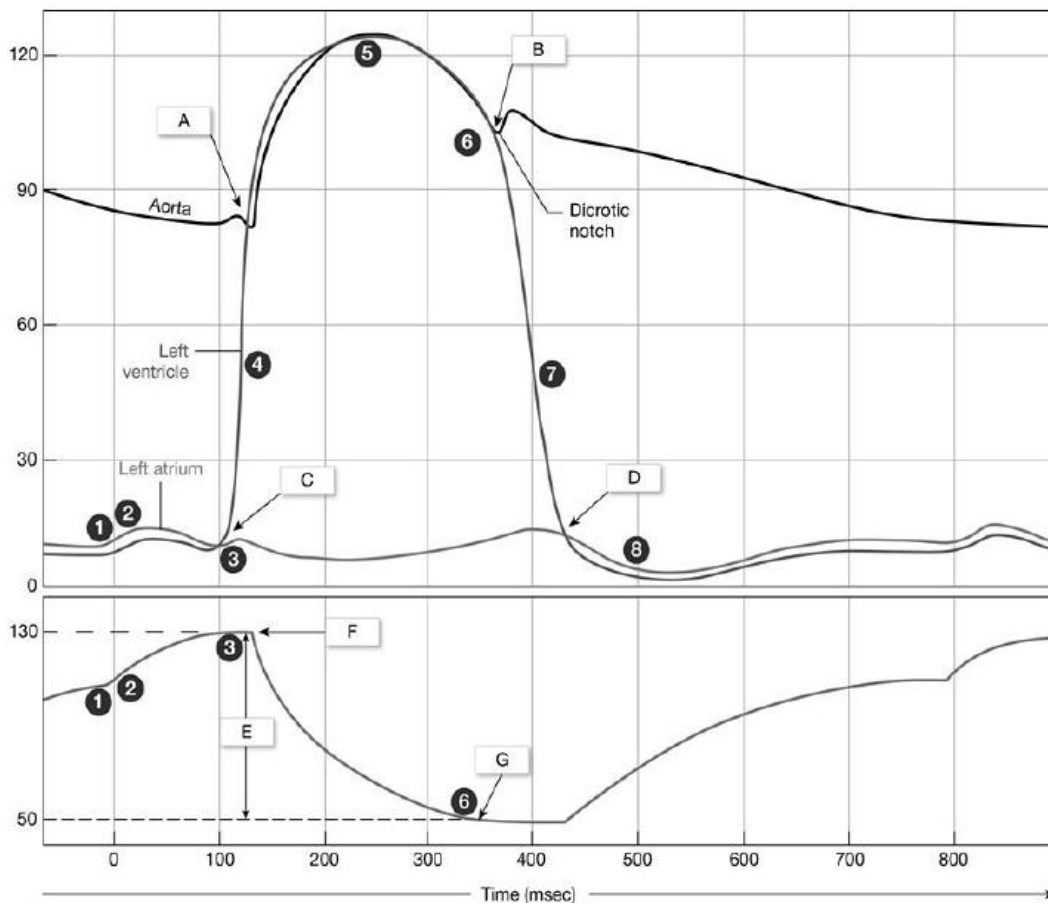
The Cardiac Cycle

Questions

Please watch the following linked video: <https://youtu.be/D0-9pj08e2A>

After watching the video, please answer the following questions:

1. Hand draw (or use software to create) a representative ECG trace. Do not, however, copy and paste a diagram that was not created by you. Label and identify all the features of your ECG trace and explain what each feature means/represents in your own words. (4 marks)
2. Describe and explain the cardiac cycle as shown in the diagram below. Your answer must be thorough and comprehensive. You should draw an ECG trace above the diagram and relate it to the events in the pressure and volume curves. In addition, you should identify when the S₁ and S₂ sounds occur and relate them to the events in the cardiac cycle (i.e., draw a phonograph at the bottom of the diagram and correlate it to the events of the cardiac cycle). You should also label the ventricular systolic and diastolic phases in the cardiac cycle diagram (at the top of the diagram). Lastly, you shall explain all the features in the the pressure and volume graphs. Use the number and alphabet labels in the diagram when explaining the phases/features of the volume and pressure graphs. (11 marks)



Changes in Heart Rate on Moving from a Squatting to a Standing Position

Questions

Please watch the following linked video: <https://youtu.be/hjAohKRbFPk>

After watching the video, please answer the following questions:

Heart Rate			
Average Heart Rate while Squatting (bpm)	Maximum Heart Rate while Standing (bpm)	Increase in Heart Rate (bpm)	% Increase in Heart Rate
70	100		

- What is the percentage increase in the heart rate? Why did the heart rate increase when the subject moved from a squatting to a standing position? You shall explain, in detail, all the physiological parameters that are expected to change during this process of moving from a squatting to standing position and explain the physiological basis of all these changes. (13 marks)
- Why do certain individuals feel “dizzy” when performing the experiment? What is this condition called? (2 marks)

Your explanations must be thorough and comprehensive. It should incorporate the cardiovascular physiological theories and principles that were discussed in class, in the textbook, and in this course. You may also incorporate other physiological principles that you researched independently.

Lung Volumes and Capacities

Questions

Please watch the following linked video: Part 1: <https://youtu.be/pNPIb0bK4d0>; Part 2: <https://youtu.be/2Cvz8e-Oyu8>

After watching the videos, please answer the following questions:

5. Hand draw (or use software to create) a representative spirometry graph. Do not, however, copy and paste a diagram that was not created by you. Label all the lung volumes and capacities. Please explain the meaning and significance of each volume and capacity. Your answers must be thorough. (5 marks)
6. Please describe three biological/physiological (but **not** pathological) factors that may affect the measured lung volumes and/or capacities of an individual. (5 marks)
7. A medical research report states: **"Doctors would not diagnose diabetes without checking blood sugar levels, or a broken bone without ordering an X-ray," says Dr. Aaron. "But for some reason, many doctors are not ordering the spirometry tests that can definitely diagnose asthma."** (<http://www.medicalnewstoday.com/articles/315305.php>)

Please explain what useful information can be extracted from lung spirometry results in the context of diagnosing certain lung diseases (e.g., COPD, asthma). You must conduct further independent research and cite the evidenced-based (and preferably peer-reviewed) literature to answer this question. (5 marks)

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Physiology Assignment – Evaluation Rubric

Question	2 marks	4 marks	8 marks	11 marks	13 marks
1	ECG trace drawn correctly with all the waves labelled.	ECG trace drawn correctly with all the waves labelled, and an explanation of what each wave represents.			
2	A very brief and incomplete explanation of the cardiac cycle.	An explanation of the cardiac cycle that does not incorporate the ECG trace nor the heart sounds.	A correct explanation of the cardiac cycle with the incorporation and correlation of the ECG trace and heart sounds. Identification of the phases of the cardiac cycle. Relating the numbers and letters in the diagram to specific events in the cardiac cycle.	Comprehensive explanation of the cardiac cycle with the incorporation and correlation of the ECG trace and heart sounds. Identification of the phases of the cardiac cycle. Relating the numbers and letters in the diagram to specific events in the cardiac cycle.	
3	A correct calculation of the increase in the heart rate.	A very superficial explanation of the phenomena of the increase in the heart rate.	A satisfactory physiological explanation of the increase in the heart rate based on the concepts covered in the cardiovascular unit in class.	A satisfactory physiological explanation of the increase in the heart rate based on the concepts covered in the cardiovascular	A thorough and comprehensive physiological explanation of the increase in the heart rate based on the concepts

				r unit in class. An explanation of the other physiological processes that would occur as a compensatory response.	covered in the cardiovascular unit in class. A good explanation of the other physiological processes that would occur as a compensatory response.
4	Identifies the name of the condition but no explanation provided. (1 mark)	An explanation of the correctly named condition is provided. (2 marks)			
5	A satisfactory spirometry graph is drawn.	A satisfactory spirometry graph is drawn with proper labels for all the volumes., and a brief explanation of the volumes and capacities.	A correct spirometry graph is drawn with proper labels for all the volumes and capacities, and a detailed explanation of the volumes and capacities. (5 marks)		
6	A list of three biological factors is provided, along with a superficial description.	A satisfactory explanation of three biological factors is provided.	A good description and explanation of three biological factors is provided. (5 marks)		
7	A satisfactory explanation of the utility of spirometry in the diagnosis of two diseases.	A thorough explanation of the utility of spirometry in the diagnosis of two diseases.	A thorough explanation of the utility of spirometry in the diagnosis and management of two diseases. (5 marks)		

Grading of APA format

Deductions from your overall assignment grade will occur for APA formatting errors to a maximum of 10% of the assignment value. An error receives the deduction the first time it is made and not repeated for the same error. APA deductions will be applied for errors in formatting, citing or references. Challenges or errors with spelling, grammar, sentence structure and flow or progression of ideas are addressed under the general grading criteria for scholarly papers provided below.

APA Mark deductions follow the rubric below for all written assignments:

2 or less APA errors: 2.5% deduction from the total grade (e.g., an assignment worth 25% of the final grade would receive a grade deduction of 0.625)

4 or less APA errors: 5% deduction (e.g., an assignment worth 25% of the final grade would receive a grade deduction of 1.25)

5 or more APA errors: 10% deduction (e.g., an assignment worth 25% of the final grade would receive a grade deduction of 2.5)